ISCC Fall 2026
Exercises on Lecture 2

Thierry Denœux

Exercise 1

We consider a random interval of the form $\overline{X} = [M - r, M + r]$, where $M \sim N(\mu, \sigma^2)$ is a normal random variable with mean μ and variance σ^2, and r is a positive constant.

1. Compute $\mathrm{Bel}_{\overline{X}}([a, b])$ and $\mathrm{Pl}_{\overline{X}}([a, b])$ for any real interval $[a, b]$. (The CDF of the standard normal distribution will be denoted by Φ).
2. Give the expression of the contour function $\mathrm{pl}_{\overline{X}}$ of $\overline{X}$. What does it become when $r = 0$? When $r \to +\infty$?
3. We now consider two independent random intervals $\overline{X}_1 = [M_1 - r_1, M_1 + r_1]$ and $\overline{X}_2 = [M_2 - r_2, M_2 + r_2]$, with $M_1 \sim N(\mu_1, \sigma_1^2)$, $M_2 \sim N(\mu_2, \sigma_2^2)$ and $r_1, r_2 > 0$. Compute the degree of conflict between $\overline{X}_1$ and $\overline{X}_2$.

Exercise 2

We consider a random interval $\overline{X} = [0, V]$, where V is a random variable with a standard uniform distribution.

1. Compute $\mathrm{Bel}_{\overline{X}}([x, y])$, $\mathrm{Pl}_{\overline{X}}([x, y])$, $Q_{\overline{X}}([x, y])$ for any interval $[x, y] \subseteq [0, 1]$, as well as the lower and upper expectations of $\overline{X}$.
2. Consider two independent random intervals $\overline{X}_1 = [0, V_1]$ and $\overline{X}_2 = [0, V_2]$ of the same form as $\overline{X}$.
   (a) What is the degree of conflict between $\overline{X}_1$ and $\overline{X}_2$?
   (b) Show that $\overline{X}_1 \oplus \overline{X}_2$ is of the form $[0, V_{12}]$, and compute the probability density function of V_{12}. What is the upper expectation of $\overline{X}_1 \oplus \overline{X}_2$?
   (c) Same questions for $n \geq 2$ independent random intervals $\overline{X}_i = [0, V_i]$, $i = 1, \ldots, n$.
3. We now consider two random intervals $\overline{X} = [0, V]$ and $\overline{Y} = [U, 1]$, where U and V have independent standard uniform distributions.
   (a) Compute the degree of conflict between $\overline{X}$ and $\overline{Y}$.
   (b) Show that $\overline{X} \oplus \overline{Y}$ is of the form $[X, Y]$. Compute $Q_{\overline{X} \oplus \overline{Y}}([x, y])$ for any interval $[x, y] \subseteq [0, 1]$, and the joint probability density function of random vector (X, Y).
   (c) Compute $\mathrm{Bel}_{\overline{X} \oplus \overline{Y}}([x, y])$ and $\mathrm{Pl}_{\overline{X} \oplus \overline{Y}}([x, y])$ for any interval $[x, y] \subseteq [0, 1]$.
   (d) Compute the lower and upper expectations of $\overline{X} \oplus \overline{Y}$.